

Static Systems

Unit 2

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Objectives

- Want to study control systems for static models
- **Static models** are the simplest possible

$$y = Gu \quad \text{where } G \text{ is just a number}$$

no dynamics, i.e. no differential equations

- We use a cruise-control problem as the driving example to show that feedback control is better than open-loop control in terms of
 - robustness
 - disturbance rejection

Car Model

- We carry this cruise control example through this section
- Static car model

$$y = Gu - Hd$$

y	mph	speed of car
r	mph	desired speed of car
u	volts	throttle motor voltage: range [0,5] volts
d	percent	uphill grade of roadway: typical range ± 5 percent
G	mph/volt	constant that relates throttle motor voltage to car speed: typically 20 ± 5
H	mph/percent	constant that relates roadway grade to speed reduction: typically 5 ± 2

- Nominal car specs
 - $G^o = 20$, $H^o = 5$
 - maximum speed $r = 100$ mph
 - roadway grade is a disturbance: not known ahead of time
 - 1 percent uphill grade of roadway causes 5 mph reduction in speed

Modeling Errors

- Actual car: $y = Gu - Hd$
- Nominal car model $y = G^o u - H^o d$
- Static modeling errors:
 - number of people in car
 - throttle effectiveness, i.e. the value of G
- Dynamic modeling errors:
 - wind resistance
 - rolling resistance at tire/roadway
 - delay in fuel lines

Design Objectives

- Perfect tracking for nominal plant model with no disturbances

$$y = r \quad \text{when } d = 0, G = G^o, H = H^o$$

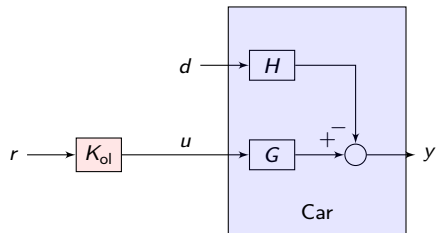
- Good disturbance rejection for nominal plant

$$y \approx r \quad \text{when } |d| \text{ is small, } G = G^o, H = H^o$$

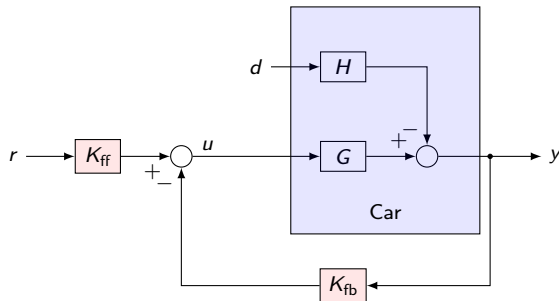
- Robust tracking and disturbance rejection

$$y \approx r \quad \text{when } |d|, |G - G^o|, |H - H^o| \text{ are small}$$

Open-loop vs Feedback Control



- throttle u set based on no information just calibration, no info feedback
- need to design one number: open-loop gain K_{ol}

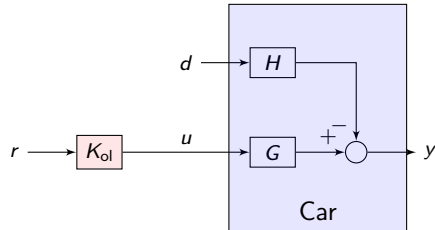


- throttle u set based on measured speed y notice the feedback loop
- need to design two numbers: feedback gain K_{fb} , feed-forward gain K_{ff}

Open-loop Controller: Design

- Actual car: $y = Gu - Hd$
Car model $y = G^o u - H^o d$
- Controller: $u = K_{ol} r$
- Controlled system:
 $y = GK_{ol} r - Hd$ (with actual car)
 $y = G^o K_{ol} r - H^o d$ (with nominal car model)
- Design: choose K_{ol} to get perfect tracking for nominal car model, no disturbance

$$r = y = G^o K_{ol} r \quad \implies \quad K_{ol} = \frac{1}{G^o} = 0.05$$



Open-loop Controller: Behavior

- Actual car: $y = Gu - Hd$
Car model $y = G^o u - H^o d$
- Controller: $u = K_{ol} r$
- Design: $K_{ol} = 0.05$
- Controlled system with open-loop controller (for nominal car model)

$$y = r - 5d$$

- Controlled system with open-loop controller (for actual car)

$$y = GK_{ol} r - Hd = 0.05Gr - Hd$$

Open-loop Controller: Assessment

- Controlled system with open-loop controller (for nominal car model)

$$y = r - 5d$$

- Controlled system with open-loop controller (for actual car)

$$y = GK_{ol}r - Hd = 0.05Gr - Hd$$

- Disturbance rejection:

- perfect tracking of reference r with no disturbance
- 1% uphill grade of hill reduces speed by 5 mph

- Robustness (with no disturbances):

- suppose model is off by 10%, so $G = 22$

$$y = 1.1r - Hd = 1.1r \quad \implies \quad \text{tracking error} = 10\%$$

- 10% model error causes 10% error in speed

Feedback Controller: Design

- Actual car: $y = Gu - Hd$
Car model $y = G^o u - H^o d$

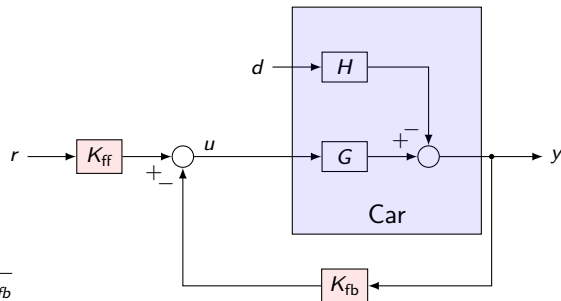
- Controller: $u = K_{ff} r - K_{fb} y$

- Controlled system (with actual car):

$$\begin{aligned} y &= Gu - Hd = GK_{ff} r - GK_{fb} y - Hd \\ &= \frac{GK_{ff}}{1 + GK_{fb}} r - \frac{H}{1 + GK_{fb}} d \end{aligned}$$

- Design: choose K_{fb} , K_{ff} to get perfect tracking for nominal car model, no disturbance

$$\begin{aligned} r = y = \frac{G^o K_{ff}}{1 + G^o K_{fb}} r &\implies 1 = \frac{G^o K_{ff}}{1 + G^o K_{fb}} \\ &\implies K_{ff} = \frac{1 + G^o K_{fb}}{G^o} \end{aligned}$$



Feedback Controller: Design ...

- Design rule: $K_{ff} = \frac{1 + G^o K_{fb}}{G^o}$ how do we pick K_{fb} ? many choices!

- Feedback or closed-loop system (with actual car):

$$y = \frac{GK_{ff}}{1 + GK_{fb}}r - \frac{H}{1 + GK_{fb}}d = \frac{1 + G^o K_{fb}}{1 + GK_{fb}} \cdot \frac{G}{G^o}r - \frac{H}{1 + GK_{fb}}d$$

- Feedback or closed-loop system (with nominal car model):

$$y = r - \frac{H^o}{1 + G^o K_{fb}}d$$

- Designing K_{fb} :
 - increasing K_{fb} improves disturbance rejection
 - lets go with $K_{fb} = 10$
 - feedback or closed-loop system (with nominal car model):

$$y = r - \frac{5}{1 + 200}d \approx r - 0.025d$$

Feedback Controller: Assessment

- Controlled system with feedback-loop controller (for nominal car model)

$$y = r - 0.025d$$

- Controlled system with feedback-loop controller (for actual car)

$$y = \frac{1 + G^o K_{fb}}{1 + GK_{fb}} \frac{G}{G^o} r - \frac{H}{1 + GK_{fb}} d$$

- Disturbance rejection:

- perfect tracking of reference r with no disturbance
- 1% uphill grade of hill reduces speed by 0.025 mph!

- Robustness (with no disturbances):

- suppose model is off by 10%, so $G = 22$

$$y = \frac{1 + G^o K_{fb}}{1 + GK_{fb}} \frac{G}{G^o} r = 1.0004r \quad \implies \quad \text{tracking error} = 0.04\%$$

- 10% model error causes 0.04 % error in speed!

Need to be careful!

- Analysis suggests that increasing feedback gain K_{fb} improves robustness and disturbance rejection
- But increasing K_{fb} too much will cause oscillations and even instability
we will see this when we use a more realistic dynamic model of the car to validate the controller
- Intuition:
 - if K_{fb} is large, we respond aggressively to measured speed y
 - if car is going too slow, controller slams the throttle
 - soon car is going too fast, and controller turns off the throttle
 - soon the car is going way too slow, and the controller slams the throttle again
 - result: oscillations
- Mathematical explanation: later!

Sensitivity Analysis

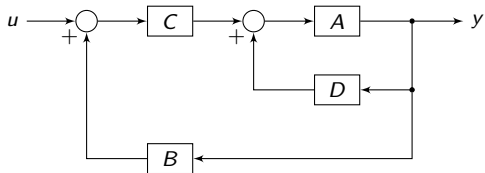
- Controlled system is affected by modeling errors in parameters G, H
- How can we study effect of small change in some model parameter?
- Sensitivity of quantity y with respect to some parameter θ

$$\begin{aligned} S_{\theta}(y) &= \frac{\text{percent change in } y}{\text{percent change in } \theta} \\ &\approx \frac{100 \cdot \Delta y / y}{100 \cdot \Delta \theta / \theta} \\ &= \frac{\partial y}{\partial \theta} \frac{\theta}{y} \end{aligned}$$

- ex: if $S_y(\theta) = 0.2$, a 3% change in the parameter θ causes a 0.6% change in the signal y

Example

Compute the sensitivity S_a of the output y with respect to changes in the parameter A .



Open-loop Control Sensitivity

- Actual car: $y = Gu - Hd$
- Open-loop Controller: $u = K_{ol}r$
- Design: $K_{ol} = 0.05$
- Controlled system with open-loop controller (for actual car, no disturbance)

$$y = GK_{ol}r - Hd = 0.05Gr$$

- How sensitive is speed of car to model parameter G around the nominal value G^o
 - calculate $S_y(G)$ evaluated at $G = G^o$

$$\begin{aligned} S_G(y) &= \frac{\partial y}{\partial G} \frac{G}{y} \\ &= 0.05r \cdot \frac{G}{0.05Gr} = 1 \end{aligned}$$

- 10% model error in G causes $10 \cdot S_y(G) = 10\%$ error in speed ... same as before!

Feedback Control Sensitivity

- Actual car: $y = Gu - Hd$
- Controller: $u = K_{ff}r - K_{fb}y$
- Feedback Controlled system (with actual car, no disturbance):

$$y = GK_{ff}r - GK_{fb}y - Hd = \frac{GK_{ff}}{1 + GK_{fb}}r$$

- Design: $K_{ol} = 0.05$, $K_{fb} = 10$
- How sensitive is speed of car to model parameter G around the nominal value G°

$$\begin{aligned} S_y(G) &= \frac{\partial y}{\partial G} \frac{G}{y} = \frac{K_{ff}(1 + GK_{fb}) - GK_{ff}K_{fb}}{(1 + GK_{fb})^2} \cdot \frac{G(1 + GK_{fb})}{GK_{ff}} \\ &= \frac{1}{1 + GK_{fb}} \Big|_{\text{nominal}} = \frac{1}{1 + 200} \approx 0.005 \end{aligned}$$

10% model error in G causes $10 \cdot S_y(G) = 0.05\%$ error in speed! ... same as before!

Summary

- Feedback control is superior to open loop control
 - better disturbance rejection
 - better robustness
- Controller: $u = K_{ff}r + K_{fb}y$
 - feed-forward gain K_{ff} designed to give perfect tracking for nominal model under no disturbances
 - feedback gain K_{fb} designed to achieve certain level of disturbance rejection
 - increasing feedback gain K_{fb} gives better disturbance rejection and robustness
 - increasing feedback gain K_{fb} too much causes oscillations and even instability